

Licong Xu

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Research Interests

My research lies at the intersection of theoretical cosmology and modern computational and statistical methods. I am interested in developing and applying state-of-the-art techniques in cosmological modelling and data analysis to extract fundamental physics from observations of the Universe. I am particularly interested in physics-based and physics-informed approaches to large-scale structure, combining theoretical modelling with machine learning and simulation-based inference. In addition, I am interested in how different forms of artificial intelligence, including machine learning, deep learning, large language models, and agentic AI, can be integrated with physical reasoning to develop new approaches to cosmological modelling, inference, and scientific discovery.

Education

University of Cambridge **Cambridge, UK**
PhD in Astronomy *Oct. 2024–Jul. 2027 (Expected)*

- Supervisor: Prof. Anthony Challinor, Dr. Boris Bolliet, Dr. Íñigo Zubeldia.
- Thesis: *Cosmological inference from the thermal Sunyaev–Zel’dovich effect and galaxy clusters.*

University of Cambridge **Cambridge, UK**
Master of Natural Sciences in Astrophysics *Oct. 2023–Jun. 2024*

- Class I (90/100).
- Supervisor: Prof. Anthony Challinor, Dr. Emilie Hertig.
- Project: *Impact of the Cosmic Microwave Background shadow on the tensor-to-scalar ratio measurements at the Simons Observatory.*

University of Cambridge **Cambridge, UK**
Bachelor of Arts in Natural Sciences (Astrophysics) *Oct. 2020–Jun. 2023*

- Class I with distinction (89.2/100).

Selected Publications

2026: Impact of non-Gaussian likelihood on cosmological constraints from the thermal Sunyaev–Zel’dovich power spectrum: a simulation-based inference analysis

L. Xu, Í. Zubeldia, J. Alvey, B. Bolliet, A. Challinor
Phys. Rev. D **114**, 043528 (2026)

2025: Open Source Planning & Control System with Language Agents for Autonomous Scientific Discovery

L. Xu, M. Sarkar, A. I. Lonappan, Í. Zubeldia, P. Villanueva-Domingo, S. Casas, et al.
ICML 2025 Workshop on Machine Learning for Astrophysics

2026: Competing with AI Scientists: Agent-Driven Approach to Astrophysics Research

T. Borrett, L. Xu, A. Nilipour, B. Bolliet, S. Pierre, E. Allys, C. Lecat, et al.
Accepted at Communications AI & Computing (Nature Portfolio)

2025: The Denario project: Deep knowledge AI agents for scientific discovery

F. Villaescusa-Navarro, B. Bolliet, P. Villanueva-Domingo, A. E. Bayer, ..., L. Xu, et al.
Accepted at Phys. Rev. X

Open Source Projects

CMBAgent — Multi-agent system for autonomous scientific research, powered by AG2

🐙 github.com/CMBAgents/cmbagent

hmfast — Differentiable JAX-native halo-model code for cosmological observables

🐙 github.com/licongxu/hmfast

Denario — Modular multi-agent system for scientific research assistance

🐙 github.com/AstroPilot-AI/Denario

Selected Talks & Presentations

- **Mar. 2026:** “Beyond AI as Assistants: Toward Autonomous Discovery in Cosmology,” *60th Rencontres de Moriond on Cosmology*, La Thuile, Italy
- **Dec. 2025:** “CMBAgent for the Fair Universe Weak Lensing Uncertainty Challenge,” *NeurIPS 2025*, San Diego, USA
- **Nov. 2025:** “Cosmological Constraints from the Thermal Sunyaev–Zel’dovich Power Spectrum — A Simulation-Based Inference Approach,” *Institute of Astronomy Wednesday Seminar*, University of Cambridge, UK
- **Sep. 2025:** “Forward modelling of thermal Sunyaev–Zel’dovich maps,” *Cambridge–LMU Cosmology Meeting*, Munich, Germany

Teaching

Lent 2026: Supervisor for Statistical Physics, University of Cambridge

Michaelmas 2025: Supervisor for General Relativity, University of Cambridge

Lent 2025: Supervisor for Introduction to Cosmology, University of Cambridge

Michaelmas 2024: Supervisor for Principles of Quantum Mechanics, University of Cambridge

Skills

AI & Agents: Multi-agent LLM systems; agent orchestration & planning

Agentic coding: Cursor, Claude Code, Codex; specification-driven development, codebase reasoning, multi-step implementation, testing, refactoring, and reusable agent workflows using skills, rules, and hooks

Personal agents: Grok Bot, OpenClaw, Hermes

Programming: Python, JAX, MATLAB, Mathematica

Machine Learning: PyTorch, TensorFlow, Scikit-learn

Scientific Computing: Simulation-Based Inference, Bayesian Inference, Monte Carlo Sampling (MCMC)

General Software: Git, Linux, $\LaTeX$, open-source development

Language: Chinese (native), English (proficient)